

Module 02: Agents — AI and Robots

Learning Objectives

1. Understand that **Robots** are Agents made of metal and code.
2. Learn about **Artificial Intelligence (AI)** (Computer brains).
3. Know the difference between a Human (Needs sleep, has feelings) and a Robot (Needs electricity, has code).

Introduction: The Robot Friend

Have you ever talked to Siri or Alexa? Are they alive? No. They are **Agents**. They have "Eyes" (Cameras), "Ears" (Microphones), and a "Brain" (Chips). They can help us, play music, and tell jokes. But they don't feel happy or sad like you do.

Key Concepts

1. The Sensor (Robot Senses)

Robots don't have skin. They have **Sensors**.

- Lidar (Laser vision).
- Heat Sensors (To find people).
- Touch Buttons. Sensors give the robot data about the world.

2. The Code (Robot Rules)

You have a brain that learns. A robot has **Code** that tells it what to do.

- "IF button is pressed, THEN say Hello."
- "IF battery is low, THEN go to sleep." Robots love rules.

3. Human vs. Machine

- Humans: Eat food, dream, love, get tired.
- Machines: Eat electricity, calculate, never get tired. We work best when we work TOGETHER (Cyborg Team!).

Activities

Activity 1: Robot Walk

Pretend to be a robot. Robots move one joint at a time.

- "Rotate head."
- "Lift arm."
- "Extend leg." Does it feel efficient? Or stiff?

Activity 2: The Turing Test (Kid Version)

One person hides behind the couch. The other person asks questions. "Do you like ice cream?" The person behind the couch tries to answer like a ROBOT. Can you trick them?

Summary

Robots are amazing helpers, but they are not people. The best future is one where Humans and Robots are friends.

References

- *The Wild Robot* by Peter Brown
- *Boy + Bot* by Ame Dyckman